

LIFEWATCH: Lifelong Wasserstein Change Point Detection

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Abstract—Change point detection methods offer a crucial capability in modern data analysis tasks characterized by evolving time series data in the form of data streams. Recent interest in lifelong learning showed the importance of acquiring knowledge and identifying new occurring tasks in a continually evolving environment. Although this setting could benefit from a timely identification of changes, existing change point detection methods are unable to recognize recurring tasks, which is a necessary condition in lifelong learning. In this paper, we attempt to fill this gap by proposing LIFEWATCH, a novel Wasserstein-based change point detection approach with memory capable of modeling multiple data distributions in a fully unsupervised manner. Our method does not only detect changes, but discriminates between changes characterized by the appearance of a new task and changes that rather describe a recurring or previously seen task. An extensive experimental evaluation involving a large number of benchmark datasets shows that LIFEWATCH outperforms state-of-the-art methods for change detection while exploiting the characterization of detected changes to correctly identify tasks occurring in complex scenarios characterized by recurrence in lifelong consolidation settings.

Index Terms—Lifelong Learning, Change Point Detection, Time-Series Data

I. INTRODUCTION

Change point detection methods provide a significant contribution for the analysis of data characterized by a dynamic and time-evolving nature. The identification of time points that define when the data distribution changes is, in fact, crucial in a variety of real-world domains characterized by multivariate time series in the form of data streams [1]. Applications include human activity recognition [2], renewable energy plant monitoring in smart grids [3], anomaly detection in geodistributed data [4], traffic flow prediction [5], and intrusion detection [6].

Change point detection methods are typically incorporated in more complex machine learning based approaches to inform changes in the data distribution and trigger appropriate actions.

In particular, lifelong learning methods are recently trying to address the challenge of acquiring new knowledge and tackling new tasks (corresponding to emerging classes, concepts, or distributions) in a continuously evolving environment. Recent methods that address this new learning paradigm are attracting increasing interest, although they are limited to image classification tasks [7]–[11]. In lifelong learning, change point detection can be used as a tool to signal changes between tasks and inform the consolidation of new tasks in memory-based approaches or, more generally, trigger updates of deep neural network-based models. However, current change point detection approaches in the literature are still not particularly suited to the needs of lifelong learning. Limitations include the inability to differentiate between changes that anticipate a new task (i.e. that never appeared before) and changes that anticipate a recurring task (i.e. the appearance of a previously seen task). As a consequence, this limitation may affect lifelong learning methods in terms of robustness and accuracy.

In this paper, we attempt to fill this gap by proposing LIFEWATCH, a novel change point detection approach that leverages memory to provide consolidation capabilities, particularly suited for the analysis of multivariate time series data in lifelong learning settings. Our method is able to efficiently identify changes in a multivariate time series in a fully unsupervised way while discriminating between changes characterized by the appearance of a new task and changes that rather describe a recurring or previously seen task. An extensive experimental evaluation on real-world multivariate time series datasets is conducted. The results of traditional change point detection, as well as lifelong consolidation, are presented and show that LIFEWATCH outperforms state-of-the-art methods for change point detection, conventional clustering, and incremental clustering in challenging experimental conditions.

The remainder of the paper is structured as follows. Section II discusses relevant change point detection works in the state-

of-the-art. Section III describes the proposed method. Section IV defines the research questions and provides details on our experiments and their setup, followed by a discussion of the results in Section V. Finally, Section VI concludes the paper, summarizing our contributions and discussing possible future work.

II. RELATED WORK

Change point detection methods can be characterized in three main types: Offline, Bayesian, and Non-Parametric [12].

Offline methods often incorporate likelihood ratio approaches that compare probability distributions of data before and after a candidate change point. The At Most One Change (AMOC) method [13] generalized this idea. Generally, these methods monitor the logarithm of the likelihood ratios within two consecutive intervals [14], [15]. Some variants introduce reducing data dimensionality as a preliminary stage [16]. As pointed out in [12], the main downside of these approaches is their reliance on known parametric models, which limits their flexibility in real-world domains. Some offline methods such as Binary Segmentation (BINSEG) [17] and Segment Neighborhoods (SEGNEIGH) [18] are capable of discovering more than one change point in a time series, by means of a greedy binary segmentation approach relying on a cost function. One alternative approach for binary segmentation is known as Wild Binary Segmentation and leverages the CUSUM statistic [19] (WBS). Additional state-of-the-art methods in this category include extensions that include pruning algorithms, resulting in a reduced execution time, such as Pruned Exact Linear Time [20] (PELT) and Robust Functional Pruning Optimal Partitioning (RFPOP) [21], which presents a higher robustness to outliers. Unfortunately, these methods are not capable of analyzing multivariate data.

Offline and online Bayesian approaches are included in probabilistic methods. Prophet [22] is a Bayesian forecasting method that supports time series with change points. Bayesian Online Change Point Detection [23] is another online approach that is based on two assumptions: *i*) a time series can be partitioned into non-overlapping states; *ii*) data points in each state are distributed independently and identically. Bayesian methods compute the run length distribution and update the relevant statistics using Bayes' theorem. Finally, the detection of change points is based on the comparison of probability values. Model Selection (BOCPDMS) [24] and spatio-temporal models (RBOCPDMS) [25] have been introduced in recent extensions.

Non-parametric change-point detection methods are usually based on tests of hypothesis, leveraging a test statistic, and a pre-defined threshold. Notable methods pertaining to this class include Kernel Change-Point Analysis (KCPA) [26], Nonparametric Change Point Detection (CPNP) [27], and Energy Change Point (ECP) [28]. Alternative non-parametric methods are distance-based. One representative example is the WATCH method [29] which exploits the Wasserstein distance metric. Instead of comparing two consecutive intervals, the method compares incoming data points to the stored part of

the current distribution and decides whether they are from the same distribution.

III. METHOD

In this section, we describe LIFEWATCH, our change point detection method designed to work in lifelong settings. Building upon our preliminary efforts in [29], the method leverages the Wasserstein distance as a non-parametric distance-based approach to compare data distributions. In the following, we first introduce the Wasserstein distance, which is a building block of the method. After that, we provide a detailed algorithm and description of the method.

A. Wasserstein distance

The Wasserstein distance is a metric defined by a probability distribution on a given metric space M . The literature provides an intuitive description of the metric considering the distribution as a unit amount of soil piled on M and the metric as the minimum "cost" of turning one pile into the other. Precisely, the metric value is considered the amount of earth to move multiplied by the mean distance it has to be moved. This analogy explains why the metric is known as the earth mover's distance.

Following the definition of the Wasserstein distance from [30], let $P(\mathbb{R}^d)$ be the set of Borel probability measures on $\mathbb{R}^d$ and $P_p(\mathbb{R}^d)$ be the subset of such measures with a finite moment of order $p \in [1, \infty)$. For $P, Q \in P(\mathbb{R}^d)$, let $\Gamma(P, Q)$ be the set of probability measures γ on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals P and Q , i.e., such that $\gamma(B \times \mathbb{R}^d) = P(B)$ and $\gamma(\mathbb{R}^d \times B) = Q(B)$ for Borel sets $B \subseteq \mathbb{R}^d$. $\text{dist}(\cdot)$ is the Euclidean norm. The p -Wasserstein distance between $P, Q \in P_p(\mathbb{R}^d)$ can be defined as:

$$W_p(P, Q) = \left(\inf_{\gamma \in \Gamma(P, Q)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \text{dist}(x - y)^p \partial \gamma(x, y) \right)^{\frac{1}{p}}. \quad (1)$$

In terms of random variables X and Y with laws P and Q , respectively, the p -Wasserstein distance is the smallest value of $\{E(\text{dist}(X - Y))^p\}^{\frac{1}{p}}$ over all possible joint distributions $\gamma \in \Gamma(P, Q)$ of (X, Y) . It is important to note that the Wasserstein distance computation is based on an approximated implementation, as the original formulation is computationally intractable. From an intuitive point of view, having a vector space $V \subset \mathbb{R}^k$, where k corresponds to the number of dimensions, it is possible to map the distributions (P, Q) to corresponding sets of k -dimensional data points $G, H \subseteq V$. In this paper, we leverage the SAG approximated computational approach to calculate the Wasserstein distance W_P as in [30]. The approximated version supports the analysis of data points rather than probability distributions that are computationally harder to handle. Consequently, in Algorithm 1, we refer to distributions as sets of data points representing their probability distribution. Another advantage of the approximated implementation is that it allows for a time-efficient distance computation on high-dimensional data. In the following text, we use $W_A \approx W_p$ to refer to the approximated version of the Wasserstein distance.

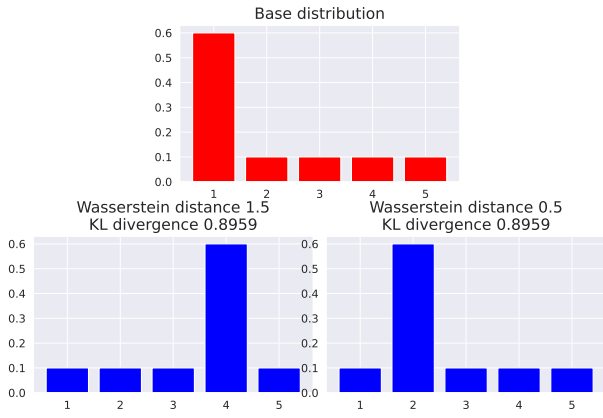


Fig. 1. Visual comparison of Wasserstein distance and Kullback-Leibler (KL) divergence. A pairwise comparison between the base distribution (in red) and two target distributions (in blue) reveals that the KL divergence estimates the same distance between the red and blue distributions. On the other hand, the Wasserstein distance reflects a better ability to measure the actual distance between the base and the two target distributions. As a result, it is able to capture a difference between the two target distributions that is uncovered by the KL divergence.

The Wasserstein distance has some unique advantages that justifies its adoption in our method. Measuring the work required to transport the probability mass between distribution states provides a meaningful and smooth representation of the distance between distributions, especially useful in domains where evaluating outcome similarity is more important than perfect likelihood matching. Figure 1 intuitively presents the informative power carried by the Wasserstein distance when compared to the Kullback-Leibler (KL) divergence. The Wasserstein distance, which measures the work required to convey the probability mass from one distribution state to the other, recognizes the difference between the red and blue distributions even though their KL divergence is the same. Motivated by these desirable properties, our method adopts the Wasserstein distance aiming at triggering a more precise and robust detection of change points. This assumption may be especially relevant in multivariate data streams, in which high-dimensionality presents increased challenges.

B. Algorithm

The input for LIFEWATCH is a time series of d -dimensional data samples $\mathbf{I} = (I_1, I_2, \dots, I_T)$. LIFEWATCH learns an initial representation of an unknown distribution D_C in an unsupervised manner. It processes the multivariate time series data stream in mini-batches B_i of length ω (sliding non-overlapping windows). Data points collected over time for the current distribution D_C are stored in a buffer. When the number of samples in the buffer reaches the desired capacity (κ parameter), D_C is added to the distribution pool P , which represents the memory of the LIFEWATCH method. The size of P can be customized to accommodate limited memory requirements. Moreover, the condition that triggers the addition of a new distribution to P can be easily controlled

to reflect domain characteristics, such as task volatility, by tuning the ϵ parameter. All distributions added to P are subsequently exploited in the change detection process. Each distribution D_j in the pool P has an assigned threshold $E[D_j]$, which is periodically updated to reflect the characteristics of the distribution. The threshold is determined as:

$$E[D_j] = \epsilon \max_{B_i \in D_j} W_A(B_i, D_j). \quad (2)$$

Intuitively, each updated threshold $E[D_j]$ conservatively reflects the largest distance computed across all mini-batches for a given distribution D_j , scaled by an ϵ factor which allows to include data points that are reasonably beyond the core of the current distribution. For each mini-batch B_i , the method determines which is the target distribution assigned. The change point detection process starts with checking whether B_i can be qualified as belonging to the current distribution D_C , comparing the Wasserstein distance between B_i and D_C with the desired threshold $E[D_C]$ for the current distribution. If the distance is lower than $E[D_C]$, we qualify B_i as belonging to D_C . On the contrary, if the distance is higher than $E[D_C]$, a change point is detected and the nature of the change point is determined. LIFEWATCH maintains two result sets for detected change points, characterizing them in new task changes (N) and recurring task changes (R).

More specifically, in order to determine whether B_i belongs to any already known distribution $D_j \neq D_C$, corresponding to a recurring task, LIFEWATCH determines the closest distribution D_k by looking for the lowest ratio between the Wasserstein distance $W_A(B_i, D_j)$ and the specific threshold for this distribution $E[D_j]$. If the distance to the closest distribution D_k is lower than $E[D_k]$, then it qualifies B_i as belonging to D_k and shifts the current distribution, i.e. D_C becomes D_k . It also adds a change point to a set R of change points for recurring tasks. On the contrary, if the distance to the closest distribution D_k is higher than $E[D_k]$, a new distribution is created: $D_C \leftarrow B_i$ and the change point is added to a set N of change points for new tasks.

After determining the proper distribution for B_i , LIFEWATCH adds B_i to D_C to let the distribution reflect the new data points. Simultaneously, it updates the corresponding threshold $E[D_C]$. The data samples for given distribution are accommodated until D_C reaches its storage capacity controlled by μ . This choice implies that the method maintains a compact version of the data stream, resulting in a lightweight memory footprint. A pseudo-code of the algorithm is presented in Algorithm 1. We emphasize some significant advantages of LIFEWATCH over existing change point detection methods:

- It works by comparing sets of points instead of working with the parameters of specific distributions, which would imply making assumptions on the distribution type;
- It characterizes change points in two types: new tasks, and recurring tasks;
- It presents the advantage of memory approaches, i.e. the opportunity to recall prior experiences, while preserving

Input: $\mathbf{I} = (I_1, I_2, \dots, I_T)$ Time series data

Result: N Change points (New Tasks),
 R Change points (Recurring Tasks).

Parameters: κ Min points required in a distrib. before starting detection,
 μ Max points stored for a distribution,
 ϵ Threshold ratio for the distance value,
 ω Mini-batch size.

```

1 Initialize pool of distributions  $P \leftarrow \emptyset$ 
2 Initialize pool of distribution thresholds  $E \leftarrow \emptyset$ 
3 Initialize current distribution  $D_C \leftarrow \emptyset$ 
4 Initialize results variables:  $N \leftarrow \emptyset, R \leftarrow \emptyset$ 
5  $\mathbf{B} \leftarrow$  Process  $\mathbf{I}$  into sequence  $(B_1, B_2, \dots, B_{\frac{T}{\omega}})$  of
   non-overlapping mini-batches  $B_i$  of size  $\omega$ 
6 for  $B_i \in \mathbf{B}$  do
7   if  $|D_C| < \kappa$  then
8      $D_C \leftarrow D_C \cup B_i$ 
9     if  $|D_C| \geq \kappa$  then
10      Determine  $E[D_C]$  from Eq. (2) with  $\epsilon$   $\triangleright$  Save
        threshold for  $D_C$ 
11       $P \leftarrow P \cup D_C$   $\triangleright$  Add  $D_C$  to the distribution pool
12    end
13  else
14     $v \leftarrow W_A(B_i, D)$   $\triangleright$  Compute Wasserstein Distance
15    if  $v > E[D_C]$  then
16       $D_k =$ 
        
$$\begin{cases} \arg \min_{D_j \in P} \frac{E[D_j]}{W_A(B_i, D_j)}, & \text{if} \\ \exists_{D_j \in P} W_A(B_i, D_j) < E[D_j] \\ \emptyset, & \text{otherwise} \end{cases}$$

17      Set change point  $c \leftarrow i * \omega$ 
18      if  $|D_k| = 0$  then
19         $N \leftarrow N \cup \{c\}$   $\triangleright$  Change point (New task)
20      else
21         $R \leftarrow R \cup \{c\}$   $\triangleright$  Change point (Recurring
        task)
22      end
23       $D_C \leftarrow D_k$   $\triangleright$  Set  $D_k$  as a current distribution
24    end
25    if  $|D_C| < \mu$  then
26       $D_C \leftarrow D_C \cup B_i$ 
27      Determine  $E[D_C]$  from Eq. 2 with  $\epsilon$   $\triangleright$  Update
        threshold for  $D_C$ 
28    end
29  end
30 end
31 return  $N, R$ 

```

Algorithm 1: Pseudo-code of the proposed LIFEWATCH method

a lightweight memory footprint;

- It automatically updates thresholds for each learned distribution without human intervention;
- It avoids unnecessary comparisons of new data points with distributions in the pool if they are recognized as belonging to the current distribution.

IV. EXPERIMENTS

The experimental study is designed to address the following research questions:

- **RQ1:** Does LIFEWATCH offer accurate and precise detection of change points on a large number of benchmark datasets and reference competitor methods?
- **RQ2:** Can the LIFEWATCH characterization of change points be effectively leveraged in a lifelong consolidation setting to accurately identify the number of tasks in scenarios with recurrence?
- **RQ3:** Is LIFEWATCH competitive when compared to a heterogeneous set of methods for change point detection, conventional clustering, and stream clustering methods in the lifelong consolidation setting?

In the following, we describe our datasets, scenarios, and metrics. Subsequently, we discuss the results obtained in our experiments.

A. Datasets

In addition to a large number of univariate and multivariate datasets (42) in the change point detection task [31], we adopted the following multivariate time series datasets to evaluate the methods with challenging data representing real-world scenarios for change point detection and lifelong consolidation tasks:

- 1) **HAR** [32] (561 features): Human Activity Recognition dataset containing sequences of 6 types of human activities such as walking, sitting, standing, and laying. In our experiments, we analyze time series data from all human participants separately and report the average values of the metrics.
- 2) **Movement** [33] (90 features): The Libras movement dataset contains sequences of 15 hand movement types in the official Brazilian signal language.
- 3) **Traffic** [34] (17 features): This dataset contains sequential urban traffic measurements. We obtain a discrete attribute with 3 classes (*low*, *medium*, and *high*) by performing discretization on the *slowness* attribute.
- 4) **Eye** [35] (14 features): Dataset containing records of EEG measurements while observing whether the eye of the subject is opened or closed.

B. Lifelong learning scenarios

In our experiments, we are interested in reproducing a lifelong learning scenario with newly appearing tasks over time. For this reason, we adopt time series datasets presented as a stream of mini-batches. We enumerate the set of possible tasks (corresponding to the k distinct classes in each dataset) and we generate 10 random permutations of them to extract a sequential scenario, whenever possible. As for the **Traffic** and **Eye** datasets (6 and 2 classes, respectively), we preserve the full set of possible task permutations. In order to further increase the complexity and realism of the scenario, we generate a sequence of tasks with a three-time recurrence, resulting in sequential scenarios of $(k \times 3)$ tasks. We argue that recurrence increases the difficulty of the experimental conditions for each considered method, which has to determine whether a change in the data distribution corresponds to a new, never seen before task, or to an already seen task. As a result,

an incorrectly identified type of change may result in wrong inferences for new data points.

C. Metrics

Our evaluation is divided in two parts: change point detection and lifelong consolidation. Each part presents different metrics described in the following:

- **Change Point Detection** – metrics adopted in the TCPD Benchmark: [31]
 - **F1**: Harmonic mean of Precision and Recall. Recall describes the ability of the model to find all relevant change points, while Precision measures how well the model can identify only relevant change points.
 - **Cover**: Describes how well the ground truth segments are covered by the predicted segments. In the time series case, a segment corresponds to a specific subsequence of contiguous points. The details on Cover metric can be found in [31].
- **Lifelong consolidation**:¹
 - **Number of Clusters**: Number of identified tasks.
 - **Homogeneity**: Assesses cluster labeling given ground truth. It has a perfect score (1.0) when each cluster has data points belonging to the same task.
 - **Completeness**: Proportion of points from each task assigned to the same cluster. It has perfect score (1.0) when all data points from the same task are into the same cluster.
 - **V-measure**: Harmonic mean of homogeneity and completeness.

V. RESULTS AND DISCUSSION

In this section, we describe the results obtained in two tasks: change point detection and lifelong consolidation.

A. Change point detection

To evaluate change point detection in non-recurring scenarios, we adopt the Turing Change Point Detection (TCPD) Benchmark [31]. It allows the evaluation of any new change point detection method on 42 datasets and compares its results to 13 pre-implemented algorithms. We extend the benchmark including the LIFEWATCH method and the datasets mentioned in Section IV-A. The evaluation is based on F1 and Cover metrics, described in Section IV-C. Results are reported in Table I, where each method is optimized via grid search using benchmark sets of parameter values, and the best results are reported on each line. The results show that our LIFEWATCH method achieves the best results in terms of average Cover and F1 metrics, in both univariate and multivariate datasets, also when compared to state-of-the-art algorithms such as Bayesian Online Change Point Detection (BOCPD) [23] and Energy Change Point (ECP) [28]. We experimentally observed that small values for the batch size parameter (e.g. 2, 4, 8) yield

the best results in terms of F1 and Cover on all datasets analyzed. This generalized behavior demonstrates the stability and accuracy of the method when the data granularity is set to a few data points. In this setting, LIFEWATCH is able to detect changes in a timely manner while avoiding false alarms. A purely incremental setting (one sample at a time) was shown to be too sensitive to false positives, whereas a large batch size (higher powers of 2) was shown to yield false negatives. Overall, these results confirm the potential of our method in a typical change point detection task, in which we are broadly interested in identifying changes, without worrying about their characterization (**RQ1**).

TABLE I
CHANGE POINT DETECTION: AVERAGE RESULTS IN TERMS OF COVER AND F1 METRICS. BLANK VALUES INDICATE THAT THE METHOD DOES NOT SUPPORT MULTIVARIATE DATA. RESULTS FOR THE BEST PERFORMING METHOD ARE MARKED IN BOLD.

	Univariate		Multivariate	
	Cover	F1	Cover	F1
AMOC	0.726	0.778		
BINSEG	0.760	0.833		
BOCPD	0.769	0.857	0.705	0.785
BOCPDMS	0.737	0.620	0.285	0.263
CPNP	0.538	0.648		
ECP	0.711	0.789	0.616	0.665
KCPA	0.619	0.679	0.583	0.671
PELT	0.706	0.766		
PROPHET	0.574	0.537		
RBOCPDMS	0.422	0.349	0.069	0.092
RFPOP	0.403	0.517		
SEGNEIGH	0.763	0.832		
WBS	0.417	0.519		
ZERO	0.573	0.658	0.268	0.332
LIFEWATCH	0.798	0.908	0.713	0.800

We assess the statistical significance of the results by adopting a rank plot for the F1 metric computed by averaging the ranks obtained by the algorithm over all datasets [36]. We apply Holm’s procedure to calculate the significant differences considering all the algorithms [37]. Figure 2 shows that LIFEWATCH strongly outperforms other methods except for BOCPD, which obtains the same rank value for multivariate datasets. If the distance between two algorithms exceeds the critical difference (CD), then the difference between them is significant. It looks clear that, from a statistical viewpoint, LIFEWATCH is on par with BOCPD, even if LIFEWATCH obtains the best average results in terms of Cover and F1 metrics. Figure 3 presents the same comparison but for univariate datasets, showing that LIFEWATCH achieves the best rank across all univariate datasets.

We compare the reported results with a variant of the method that leverages the more popular KL-divergence. As expected, the extracted results appear much worse than those extracted using the Wasserstein distance: 0.510 Cover and 0.568 F1 for univariate datasets; 0.356 Cover and 0.432 F1 for multivariate datasets. Using the Wasserstein distance yields an improvement of 0.288 in terms of Cover and 0.340 in terms of F1 for univariate datasets. Moreover, it yields an improvement

¹We adopted clustering metrics due to the unsupervised and rather model-less nature of our method, which does not allow us to calculate metrics such as *Forward Transfer* adopted in the lifelong learning.

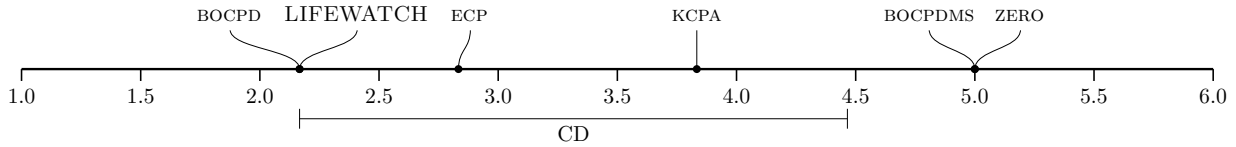


Fig. 2. Change point detection: F1 rank plot for all multivariate datasets analyzed.

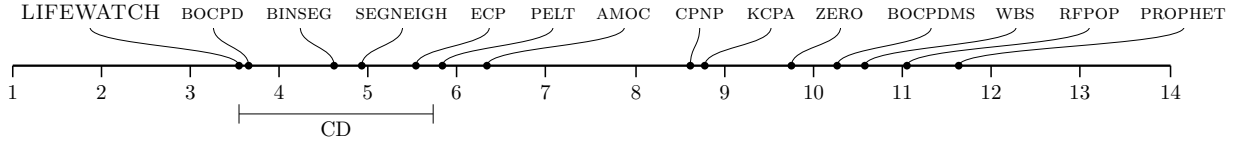


Fig. 3. Change point detection: F1 rank plot for all univariate datasets analyzed.

of 0.357 in terms of Cover and 0.232 in terms of F1 for multivariate datasets. This evaluation confirms our hypothesis about the Wasserstein distance being a more suitable and powerful metric for change point detection in the experimental setting analyzed in this study.

B. Lifelong consolidation

These experiments are focused on assessing the potential of our method in correctly identifying tasks in a consolidation experiment, leveraging its ability to differentiate between two types of change points as a feedback in a lifelong learning setting. We compare our results with three classes of methods:

- **Change point detectors:** We devise a simple procedure to create a clustering model, in order to compare LIFEWATCH with other change point detectors on a common ground: when a change point is detected, we create a new cluster and assign data points from the current mini-batch to it. Otherwise, we assign them to the closest cluster in the model. We selected BOCPD [23] and ECP [28] since they showed to be the best two performing competitor methods in the change point detection task on our datasets (see Table I).
- **Conventional clustering:** We consider three methods, each of them representative of different clustering paradigms: hierarchical (BIRCH [38]), density-based (OPTICS [39]) and partitional (MEAN-SHIFT [40]). Since these methods are not incremental, we execute experiments providing them with the full dataset as input, thus giving them a strong advantage for the identification of the target number of clusters.
- **Stream clustering:** We consider state-of-the-art methods in incremental/stream clustering: DENSTREAM [41], HDBSCAN [42], and TRESTLE [43]. In order to setup a good value for their parameters (variants of the *eps*, *minPts* parameters in the well-known DBSCAN algorithm), we apply the Elbow rule [44] considering the distances on a first batch of data consisting of 100 points.

Results in Table II show that LIFEWATCH presents the best results in terms of the number of identified clusters (NC) with respect to the actual number of tasks in the scenarios

for all datasets except for **Eye**. A first inspection shows that MEAN-SHIFT extracts, on average, a number of clusters that is closer to the ground truth of that specific dataset when compared to LIFEWATCH (difference of 0.5). However, a more detailed analysis reveals that MEAN-SHIFT mostly extracts the same number of clusters for all datasets. It is also important to remark that clustering algorithms have the advantage of seeing the full datasets, whereas LIFEWATCH accomplishes this task in an incremental way, without any prior knowledge (RQ2). Moreover, the average number of clusters extracted by LIFEWATCH is very close to the optimal results ($2.50 \approx 2$), which is a remarkable result. A visual perspective of lifelong consolidation results is shown in Figure 4, which allows to easily grasp the best performing methods (in terms of number of identified tasks) by comparing bars with the dashed line (number of actual tasks).

Overall, these results show that, even if LIFEWATCH is not strictly a consolidation method resembling prior attempts in lifelong learning, it can be fruitfully exploited for this task, without the burden of training complex deep neural networks (RQ2). Moreover, results show that other change point detectors are not able to provide high-quality results for this task, since they can only detect generic changes and cannot detect a recurrence of tasks due to their lack of memory. Finally, while LIFEWATCH identifies a near-optimal number of tasks, classical clustering algorithms and stream clustering algorithms are not able to extract a precise representation of the data in the lifelong scenario (RQ3).

VI. CONCLUSIONS AND FUTURE WORK

In this work, we have tackled a challenging and previously unaddressed problem of lifelong change point detection with multivariate time series data in the form of data streams. We proposed LIFEWATCH, a novel Wasserstein distance-based change point detection approach with memory capable of modeling multiple data distributions in a fully unsupervised manner, and discriminating between changes characterized by the appearance of a new task and changes that rather describe a recurring or previously seen task. An extensive experimental evaluation with a large number of benchmark datasets reveals

TABLE II

LIFELONG CONSOLIDATION: RESULTS IN TERMS OF *Homogeneity* (H), *Completeness* (C), *V-Measure* (VM), AND *Number of Clusters* (NC) WITH FOUR MULTIVARIATE TIME SERIES DATASETS. THE ACTUAL NUMBER OF TASKS FOR EACH DATASET (GROUND TRUTH) IS REPORTED IN PARENTHESIS. THREE CLASSES OF METHODS: *i*) CHANGE POINT DETECTORS COMBINED WITH A UNIFIED CLUSTERING STRATEGY (TOP), *ii*) CONVENTIONAL CLUSTERING (CENTER), AND *iii*) STREAM CLUSTERING (BOTTOM). THE BEST RESULTS IN TERMS OF NC ARE MARKED IN BOLD.

	Movement				HAR			
	H	C	VM	NC (15)	H	C	VM	NC (6)
LIFEWATCH	0.26±0.05	0.30±0.02	0.28±0.04	15.20±5.53	0.42±0.05	0.60±0.14	0.48±0.04	6.40±3.32
BOCPD	0.41±0.04	0.33±0.02	0.37±0.03	60.30±2.15	0.55±0.07	0.40±0.04	0.46±0.05	20.40±2.15
ECP	0.21±0.04	0.25±0.04	0.23±0.04	19.30±1.49	0.56±0.05	0.42±0.04	0.48±0.04	16.20±0.87
BIRCH	0.75±0.01	0.56±0.01	0.64±0.01	46.60±4.39	0.93±0.02	0.45±0.03	0.61±0.03	87.20±12.20
MEAN-SHIFT	0.00±0.00	1.00±0.00	0.00±0.00	1.00±0.00	0.39±0.00	0.98±0.02	0.55±0.01	2.80±0.87
OPTICS	0.41±0.01	0.51±0.00	0.46±0.01	29.50±0.67	0.17±0.03	0.37±0.01	0.23±0.03	10.50±1.69
DENSTREAM	0.42±0.24	0.23±0.13	0.30±0.17	4.30±0.64	0.64±0.17	0.44±0.15	0.52±0.15	4.40±1.50
HDBSCAN	0.78±0.21	0.39±0.12	0.52±0.15	3.80±0.40	0.90±0.14	0.71±0.34	0.76±0.28	4.10±3.39
TRESTLE	0.25±0.02	0.55±0.04	0.35±0.02	3.70±0.46	0.38±0.01	0.97±0.03	0.54±0.02	2.00±0.00
	Traffic				Eye			
	H	C	V	NC (3)	H	C	V	NC (2)
LIFEWATCH	0.16±0.13	0.25±0.21	0.19±0.16	3.17±1.34	0.11±0.04	0.10±0.04	0.10±0.04	2.50±0.50
BOCPD	0.26±0.05	0.16±0.02	0.20±0.03	13.00±1.53	0.77±0.00	0.22±0.02	0.35±0.03	15.50±2.50
ECP	0.19±0.06	0.16±0.06	0.17±0.06	7.00±1.29	0.72±0.05	0.25±0.03	0.37±0.04	9.00±0.00
BIRCH	0.61±0.02	0.25±0.01	0.36±0.01	21.17±0.69	0.85±0.05	0.30±0.01	0.44±0.01	7.50±0.50
MEAN-SHIFT	0.00±0.00	1.00±0.00	0.00±0.00	1.00±0.00	0.17±0.00	0.16±0.00	0.17±0.00	2.00±0.00
OPTICS	0.70±0.00	0.28±0.00	0.40±0.00	20.00±0.00	0.58±0.00	0.18±0.00	0.27±0.00	16.00±0.00
DENSTREAM	0.30±0.30	0.50±0.37	0.21±0.20	3.17±2.11	0.10±0.01	0.19±0.01	0.13±0.00	4.00±1.00
HDBSCAN	0.69±0.44	0.20±0.13	0.31±0.20	9.17±1.77	0.90±0.00	0.32±0.00	0.47±0.00	9.00±0.00
TRESTLE	0.07±0.04	0.13±0.02	0.09±0.03	2.17±0.37	0.64±0.12	0.40±0.07	0.49±0.09	3.00±0.00

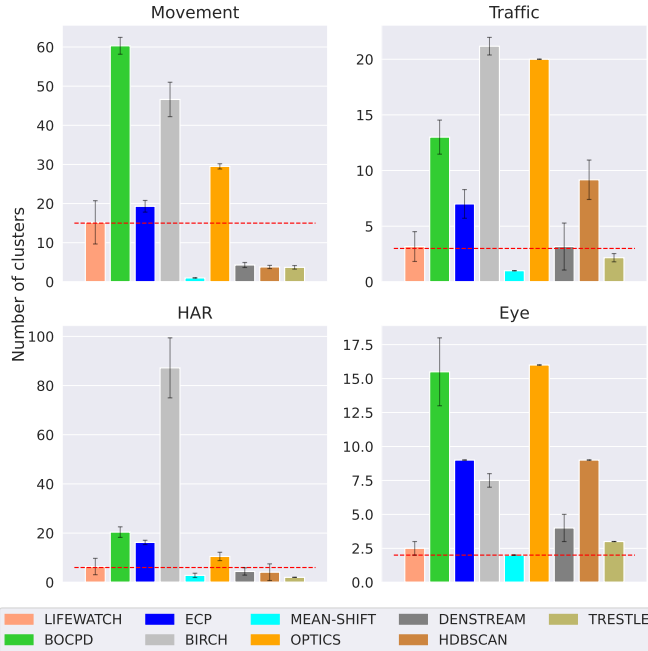


Fig. 4. Lifelong consolidation: Number of clusters extracted (average and standard deviation) for all methods and datasets considered. The dashed line is the ground truth, i.e. the number of actual tasks in each dataset.

that our method is capable of accurately and precisely detecting changes in multivariate time series data, outperforming all considered change point detection competitor methods. Additional results reveal the potential of our method when exploiting the characterization of detected changes in two different types to use in lifelong consolidation. Our method outperforms all considered change point detection, clustering, and stream clustering competitor methods considered in the identification of the correct number of tasks in complex lifelong learning scenarios characterized by recurrence. As future work, we aim to investigate extensions of our method to support lifelong learning settings, as well as improvements in the change point detection procedure to identify periodic behavior in time series data. Moreover, we envision the opportunity to investigate extensions including a systematic evaluation and compression of the learned distributions, which could overcome possible limitations of the method in fast evolving scenarios with a large number of distributions.

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